

Carmen Pelayo Fernández

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PROFILE

Research analyst at BBVA Research (Big Data & AI Economic Analysis), with over two years applying big data, NLP, large language models and econometrics to macroeconomic, geopolitical and financial questions, with a strong interest in the ethics and responsible governance of AI (taught as a teaching assistant at Boston University). MSc in Data Science (Boston University) and BSc in Management & Technology with a minor in Economics (UC3M). Advanced in Python and R, building economic, financial and econometric models on large, unstructured datasets. Native Spanish; C2 English; A1 German.

PROFESSIONAL EXPERIENCE

BBVA Research — Research Analyst, Big Data & AI Economic Analysis *Madrid, Spain | Jun 2024 – Present*

- Built worldwide big-data indicators of Economic and Trade Policy Uncertainty (EPU/TPU) from GDELT, extending Baker & Bloom and Caldara et al. from a handful of national outlets to global press coverage.
- Benchmarking foundation time-series models (Chronos-2, Bistro and others) against classical baselines (SARIMA, linear regression) for short-term economic forecasting.
- Developing a novel measure of European defence-industry integration by scraping and cleaning defence-sector news and scoring inter-firm collaboration with an LLM-based tool (GABRIEL).
- Tracked real-time candidate sentiment in the 2026 Colombian elections through text mining of news, social-media and polling big data.
- Estimated the macroeconomic impact of semiconductor supply-and-demand shocks using Local Projections.

Boston University — Teaching Assistant, AI Ethics (DS380) *Boston, MA | Jan – May 2024*

DS380 (Data, Society & Ethics) covers ethical theories, algorithmic bias, privacy and security, AI law (incl. the EU AI Act), intellectual property and the societal impact of AI.

- Led discussion sections of 30 students, prepared case studies and held office hours; received the 2023–24 Graduate Teaching Excellence Award.

Technology & Strategy Consulting — Avasant; Deloitte *Madrid, Spain | Sep 2022 – Sep 2023*

Associate Consultant, Avasant (Apr–Sep 2023) · Enterprise Technology Analyst, Deloitte (Sep 2022–Apr 2023)

- Advised on enterprise strategic transformation — scenario analysis for a multi-million IT-offshoring contract (US healthcare) and vendor negotiations for a UK vaccine R&D NGO — and supported a post-merger Salesforce consolidation.

PUBLICATIONS

- Pelayo, C. (2025). [Measuring Technological Progress in Real Time with ArXiv](#) — BBVA Research.
- Pelayo, C. (forthcoming, 2026). *From Macro Fundamentals to Digital Footprints: Elevating NPL Nowcasting with Big Data and Ensemble Learning* — BBVA Research.

EDUCATION

Boston University — MSc Data Science *Boston, MA | Sep 2023 – May 2024*

GPA 3.7/4.0; CDS Merit Scholarship and Graduate Teaching Excellence Award. Thesis: short-term demand forecasting with neural-network architectures (NeuralProphet, N-BEATS vs. ARIMA / Holt-Winters).

Universidad Carlos III de Madrid — BSc Management & Technology, Minor in Economics *Madrid, Spain | Sep 2018 – Jul 2022*

Grade 8.4/10; honours in Environmental Economics. Bachelor thesis (10/10, with Honours): dynamic location-recommendation system for technology firms in Europe.

University of Wisconsin–Madison — Academic Exchange (Information Systems) *Madison, WI | Sep 2021 – May 2022*

Coursework: Microeconomic & Macroeconomic Theory, Quantitative Tools for Economics, Machine Learning for Business Analytics.

SELECTED PROJECTS

- [ECB Nowcasting Toolbox \(Python\)](#) — Open-source Python implementation of the European Central Bank's public Nowcasting Toolbox.
- MIT Energy & Climate Hackathon** (Finalist, 2023) — AutoML ensemble forecasting building energy use for Schneider Electric; $R^2 = 0.83$ on >1.4M records.

SKILLS & LANGUAGES

Programming & tools: Python, R, SQL, Git, AWS, Hugging Face, PyTorch, GDELT. **Methods:** Econometrics, Local Projections, time-series & foundation-model forecasting, machine & deep learning, NLP, LLMs, text mining, big-data pipelines. **Languages:** Spanish (native), English (C2), German (A1).

AWARDS & CERTIFICATIONS

Graduate Teaching Excellence Award, Boston University (2024) · CDS Merit Scholarship, Boston University (2023) · MIT Energy & Climate Hackathon Finalist (2023) · CEMFI Summer School — Data Science for Economics (2025) · Salesforce Certified Administrator (2023).